

# FICTIONAL INDUSTRIAL TRAINING COMPLEX

## Training Example – Standard Operating Procedure & Incident Review

|                     |                                                                          |
|---------------------|--------------------------------------------------------------------------|
| Document ID         | SOP-SRU-801-REV0 (Training Example)                                      |
| Title               | Emergency Isolation Procedure for Sulfur Recovery Knockout Drum (KD-801) |
| Facility            | Sulfur Recovery Training Unit                                            |
| Effective Date      | January 15, 2025                                                         |
| Reference Standards | API 521, ISO 45001:2018 (Illustrative Only)                              |

### Purpose

This is a fictional engineering training document intended to demonstrate professional SOP formatting. It does not represent a real facility or operating procedure.

### Equipment Information

| Parameter                 | Value                                |
|---------------------------|--------------------------------------|
| Primary Asset             | KD-801 Sulfur Recovery Knockout Drum |
| Associated Blower         | B-801A Process Gas Blower            |
| Normal Pressure           | 9.8 bar                              |
| MAWP                      | 15.0 bar                             |
| Operating Temperature     | 210 °C                               |
| High Pressure Alarm       | 13.5 bar                             |
| Relief Valve              | PSV-801A (14.2 bar)                  |
| Emergency Isolation Valve | EIV-801                              |

### Illustrative Operating Procedure

- Verify work authorization, instrument readiness, and communications.
- Purge vessel with inert nitrogen before introducing process gas.
- Start auxiliary systems and bring blower B-801A online gradually.
- Maintain drum level within operating band and observe pressure trend.
- If pressure approaches alarm level, reduce upstream flow and prepare isolation.
- Record all operating parameters in the shift log.

### Illustrative Lockout / Tagout

- Stop blower B-801A and isolate electrical supply.
- Close inlet and outlet isolation valves.

- Apply locks, tags, and verify zero energy.
- Drain and vent equipment before maintenance.

## **Illustrative Incident Review**

**Incident ID:** TRN-2025-031

**Scenario:** Simulated increase in knockout drum pressure during a training exercise following reduced downstream gas flow.

**Illustrative Root Cause:** Partial fouling of the downstream gas filter increased flow resistance, resulting in elevated vessel pressure.

**Illustrative Corrective Actions:** Replace filter element, review differential-pressure trends, improve inspection frequency, and conduct refresher operator training.

## **Document Notice**

This document is entirely fictional and is intended solely for educational, formatting, and training purposes.